



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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Course: Fundamentals of Data Science Course Code: DSA0404

Slot: A

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Data Interpretation

1. Data Summarization

Given salaries (₹'000):

32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120

(a) Mean and Median

Mean = 52.82 ₹'000

Median = 42 ₹'000

(b) Why is mean poor here?

The values 95 and 120 are very high compared with most salaries. They pull the mean upward from the typical salary level.

(c) Better measure of dispersion

IQR is more trustworthy because it is resistant to extreme outliers, while standard deviation is strongly affected by them.

2. Outlier Treatment

Data:

32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120

Using the $1.5 \times \text{IQR}$ rule:

$Q1 = 37$

$$Q3 = 49$$

$$IQR = Q3 - Q1 = 49 - 37 = 12$$

$$\text{Lower fence} = 37 - 1.5(12) = 19$$

$$\text{Upper fence} = 49 + 1.5(12) = 67$$

Outliers

Values below 19 or above 67 are outliers.

95 → Outlier

120 → Outlier

Cap both outliers at the upper fence of 67:

32,35,36,38,40,42,45,48,50,67,67

The mean changes from approximately 52.82 to 45.45 ₹'000.

So, the mean decreases substantially.

3. Measuring Asymmetry

Both datasets have:

$$\text{Mean} = 50$$

$$\text{SD} = 10$$

Dataset A

$$\text{Pearson skewness} = +1.2$$

Positively skewed

Has a long right tail

Shape:



Dataset B

$$\text{Pearson skewness} = -0.8$$

Negatively skewed

Has a long left tail

Shape:



Salary-like distribution

Dataset A is more likely because salaries commonly have a long right tail caused by a small number of very high earners.

4. Coefficient of Skewness

Given:

Mean = 62

Median = 58

Mode = 52

SD = 12

(a) Karl Pearson — Mode Method

$Sk = \frac{SD}{Mean - Mode} = \frac{12}{62 - 52} = 1.2$

(b) Median Method

$Sk = \frac{3 \times SD}{Median - Mode} = \frac{3 \times 12}{62 - 58} = 9$

Comment

Both indicate positive skewness. They differ slightly because the two formulas use different measures of central tendency.

5. Variance vs Standard Deviation

Line	Mean	Variance	SD	CV
A	200	64	8	4%
B	150	36	6	4%

Coefficient of Variation:

$CV = \frac{SD}{Mean} \times 100$

Line A

$$CV=2008 \times 100=4\%$$

Line B

$$CV=1506 \times 100=4\%$$

Answer

Both lines have equal relative variability (4%).

6. Sampling Distribution

Given:

$$\mu = 500$$

$$\sigma = 80$$

$$n = 64$$

(a) Standard Error

$$SE=\frac{\sigma}{\sqrt{n}} = \frac{80}{\sqrt{64}} = 10$$

(b) Probability that sample mean exceeds 520

$$Z=\frac{520-500}{10}=2 \quad P(Z>2)\approx 0.0228$$

So probability $\approx 2.28\%$.

(c) If $n = 256$

$$SE=\frac{80}{\sqrt{256}} = \frac{80}{16} = 5$$

The standard error becomes half.

This demonstrates the Central Limit Theorem / square-root law of sampling: increasing sample size reduces sampling variability.

7. Covariance Calculation

Given:

$$X = 10, 20, 30, 40, 50$$

$$Y = 15, 18, 25, 30, 42$$

Means:

$$\bar{X} = 30$$

$$\bar{Y} = 26$$

Using population covariance:

$$\text{Cov}(X, Y) = n \sum (X - \bar{X})(Y - \bar{Y}) \quad \text{Cov}(X, Y) = 132$$

Interpretation

The covariance is positive, so advertising spend and sales tend to increase together.

Limitation

The magnitude 132 alone does not tell us how strong the relationship is, because covariance depends on the units and scales of both variables.

Correlation standardizes the relationship and gives a value between -1 and $+1$.

8. Correlation vs Causation

Given:

$$r = 0.85$$

A high correlation does not mean ice cream causes drowning.

Statistical flaw

The newspaper is confusing correlation with causation.

Likely confounding variable

Hot weather / temperature.

During hot weather:

More people buy ice cream.

More people go swimming.

More swimming can lead to more drowning incidents.

To establish causation

Additional evidence would be required, such as:

Controlled experiments where possible

Longitudinal evidence

Controlling for temperature and other confounders

A plausible causal mechanism

Consistent results across different settings

9. Effect of Outliers on Correlation

Initially:

$r=0.42$

After removing one extreme outlier:

$r=0.89$

Interpretation

This shows that Pearson's correlation can be highly sensitive to influential outliers.

The outlier was substantially affecting the relationship.

Spearman correlation

Yes, Spearman's rank correlation is generally more robust because it uses ranks rather than raw numerical distances.

However, an extreme observation can still affect Spearman's correlation somewhat if its rank is influential.

10. Missing Values

15% of Income values are missing, and missingness is more common among unemployed respondents.

(a) Classification

This is most appropriately MAR (Missing at Random) if employment status is observed and explains the missingness.

For example:

Missing income depends on employment status, which is known.

(b) Why mean imputation causes bias

Unemployed respondents may have systematically different/lower incomes. Replacing their missing values with the overall mean can overestimate their income and distort the distribution.

(c) Better strategy

Use:

Multiple imputation

Regression-based imputation

Imputation separately by employment group

Multiple imputation is generally preferable for statistical analysis.

11. Standardization / Normalization

Age:

Range = 18–75

Income:

Range = ₹20,000–₹20,00,000

For a distance-based algorithm, income has a much larger numerical scale than age.

For example, a difference of ₹1,00,000 can dominate the distance calculation compared with a difference of a few years in age.

Therefore, without scaling, the algorithm may treat income as much more important simply because of its units.

Z-score for Age = 45

Given:

Mean = 40

SD = 12

$Z = \frac{45 - 40}{12} = 0.417$

So Age 45 is about 0.42 standard deviations above the mean.

12. Distribution Shape from Summary Statistics

Given:

Mean = 45

Median = 45

Mode = 45

Skewness ≈ 0

Kurtosis = 5.8

Mean = Median = Mode and skewness ≈ 0 indicate an approximately symmetric distribution.

However, kurtosis = 5.8 indicates a leptokurtic distribution.

This means the distribution has:

Relatively heavier tails

More extreme observations

Greater concentration around the center compared with a normal distribution, depending on the kurtosis convention used

Therefore, symmetry does not necessarily mean normality.

13. Grouped Data — Mean and Variance

Class	Midpoint x	Frequency f
0–10	5	5
10–20	15	8
20–30	25	15
30–40	35	12
40–50	45	10

Total:

$N=50$

Take assumed mean:

$A=25$

Class width:

$h=10$

Step deviation:

$$d' = hx - A$$

x	f	d'	fd'	fd' ²
5	5	-2	-10	20
15	8	-1	-8	8
25	15	0	0	0
35	12	1	12	12
45	10	2	20	40
Total	50		14	80

Mean

$$\bar{x} = A + hN \sum fd' = 25 + 10(5014) \quad \bar{x} = 27.8$$

Variance

$$\sigma^2 = h^2 [N \sum fd'^2 - (\sum fd')^2] = 100 [5080 - (14/50)^2] \quad \sigma^2 = 152.16$$

14. Comparing Two Distributions via CV

Portfolio A

$$\text{Mean} = 12\%$$

$$\text{SD} = 3\%$$

$$\text{CV} = 123 \times 100 = 25\%$$

Portfolio B

$$\text{Mean} = 18\%$$

$$\text{SD} = 5\%$$

$$\text{CV} = 185 \times 100 = 27.78\%$$

Conclusion

Portfolio B has:

Higher absolute risk: SD = 5% vs 3%

Higher relative risk: CV = 27.78% vs 25%

Therefore, the statement is partly correct: B has greater absolute risk, but it is also slightly riskier per unit of return according to CV.

15. Outlier's Effect on Variance

Original:

$$n = 10$$

$$\text{Mean} = 42$$

$$\text{Variance} = 25$$

$$\text{One value} = 40$$

The value 40 is close to the mean.

It is incorrectly changed to 400.

(a) Effect on Mean

Original total:

$$42 \times 10 = 420$$

New total:

$$420 - 40 + 400 = 780$$

New mean:

$$780 / 10 = 78$$

So the mean increases from 42 to 78.

(b) Effect on Variance

Variance increases dramatically.

Why?

Variance uses squared deviations.

The incorrect value changes from:

$$40 - 42 = -2$$

to approximately:

$$400 - 78 = 322$$

Squaring 322 produces a very large value.

Therefore, variance is affected disproportionately more than the mean because large deviations are squared.

16. Continuous Distribution — Area Interpretation

Given:

$$\mu = 100$$

$$\sigma = 15$$

(a) Proportion between 85 and 130

For 85:

$$Z = \frac{85 - 100}{15} = -1$$

For 130:

$$Z = \frac{130 - 100}{15} = 2$$

Therefore:

$$P(-1 < Z < 2) = 0.9772 - 0.1587 = 0.8185$$

So approximately 81.85% of values lie between 85 and 130.

(b) Value = 145

$$Z = \frac{145 - 100}{15} = 3$$

So it is 3 standard deviations above the mean.

Under the empirical rule, ± 3 SD contains approximately 99.7% of observations.

Therefore, 145 is a potential outlier / unusually high observation.

17. Sampling Bias vs Sampling Error

Sample mean:

₹18,000

Population mean:

₹14,500

(a) Explanation

This is more likely sampling bias because the survey only uses online panel respondents.

Online respondents may systematically differ from the entire population.

(b) Will increasing sample size fix it?

No.

Increasing the sample size reduces sampling error, but it does not eliminate systematic sampling bias.

A very large biased sample can still produce a biased estimate.

18. Covariance Sign vs Correlation Strength

Given:

Pair 1: Cov = 850

Pair 2: Cov = 12

Both are positive.

Therefore, both relationships have a positive direction.

But we cannot say Pair 1 has a stronger relationship.

Why?

Covariance depends on the units and scale of the variables.

The missing information is the standard deviations of both variables.

Correlation standardizes covariance:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Thus correlation allows meaningful comparison of relationship strength.

19. Confidence Interval

Given:

$n = 100$

Sample mean = 250

SD = 40

95% confidence level

Standard error:

$SE = 100 \cdot 40 = 4$

95% margin of error:

$1.96(4) = 7.84$

Therefore:

250 ± 7.84 (242.16, 257.84)

Interpretation

The interval gives a plausible range for the population mean, based on the sample and the chosen confidence procedure.

It does not mean there is a 95% probability that the already-fixed population mean is inside this particular interval. The 95% refers to the long-run performance of the confidence-interval procedure.

20. Integrated Case Study

Daily sales (₹'000):

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

(a) Identify the Outlier

IQR Method

Sorted data:

117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500

Using the median-of-halves method:

$Q1 = 120$

$Q3 = 128$

$IQR = 128 - 120 = 8$

Upper fence:

$128 + 1.5(8) = 140$

Lower fence:

$$120 - 1.5(8) = 108$$

Therefore:

500 is an outlier

Z-score Method

Mean with outlier:

$$\bar{x} = 148.47$$

Population SD ≈ 94.03 .

For 500:

$$Z = \frac{500 - 148.47}{94.03} \approx 3.74$$

Since $|Z| > 3$, 500 is a strong potential outlier.

Conclusion

Both IQR and Z-score methods identify 500 as an outlier.

(b) Mean and Median With and Without Outlier

With 500

$$\text{Mean} = 148.47$$

$$\text{Median} = 124$$

Without 500

$$\text{Mean} = 123.36$$

$$\text{Median} = 123.5$$

Measure	With 500	Without 500
Mean	148.47	123.36
Median	124	123.5

Interpretation

The mean changes by about 25.1 ₹'000, while the median changes by only 0.5 ₹'000.

Therefore, the mean is much more sensitive to the outlier, while the median is robust.

(c) Skewness

Using the moment coefficient of skewness:

$Sk = \frac{\text{SD}^3 \text{third central moment}}{\text{SD}^3}$

The skewness is approximately:

3.46

Since skewness is strongly positive:

Distribution is highly positively/right skewed

The 500 ₹'000 observation creates a long right tail.

(d) Recommendation

The company should investigate the 500 value before reporting performance.

Evidence:

IQR method identifies 500 as an outlier.

Z-score ≈ 3.74 confirms it is unusually extreme.

Mean increases from 123.36 to 148.47 when 500 is included.

Median remains almost unchanged.

Skewness = +3.46, showing strong right skewness.